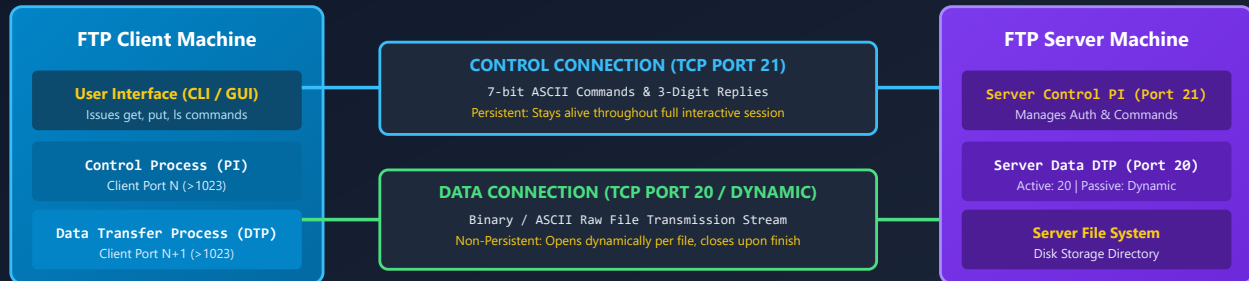


Module 07: File Transfer Protocol (FTP) Architecture

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FILE TRANSFER PROTOCOL (FTP) DUAL-CONNECTION ARCHITECTURE

Separation of Control (Port 21) & Data (Port 20), Active Mode (PORT) vs Passive Mode (PASV)



1. Why Dual Connections?

Separation of Concerns: Control uses simple 7-bit ASCII lines; Data transfers raw bytes. • **Interactive Control:** User can abort a huge 10GB file transfer mid-way via Port 21. • Prevents command choking during file transfer. 2. Active FTP Mode (PORT) • Client connects to Server Port 21. • Client sends PORT IP,p1,p2 command. • **Server initiates data connection FROM Port 20 TO Client's Port N+1.** • **Firewall Problem:** Client NAT blocks server's incoming SYN! 3. Passive FTP Mode (PASV) • Client sends PASV command to Server. • Server opens random port P (>1023) and replies: 227 Entering Passive Mode (IP,p1,p2) . • **Client initiates data connection TO Server.** • **Firewall Friendly:** Outbound connections allowed!

1. FTP Dual-Connection Architecture

File Transfer Protocol establishes two discrete logical connections over TCP: an interactive Control Connection and an ephemeral Data Connection.

Functional Connection Roles

Control Connection (Port 21): Stays persistently open throughout session; transmits 7-bit ASCII commands and 3-digit replies.

Data Connection (Port 20 / PASV): Opens dynamically per file transfer / directory listing; closes automatically upon completion.

2. Active Mode vs Passive Mode FTP

Criterion	Active Mode (PORT)	Passive Mode (PASV)
Data Connection Initiator	Server initiates connection to Client	Client initiates connection to Server
Server Data Port	Fixed Well-Known Port 20	Ephemeral dynamic unreserved port (>1023)
Client Firewall Traversal	Fails: Client NAT firewall blocks unsolicited server SYN	Succeeds: Outbound connection permitted by firewall state tables

3. Key FTP Commands

- USER / PASS :** Client authentication credentials.
- PORT / PASV :** Selects active vs passive data connection negotiation mode.
- RETR / STOR :** Retrieves file from server / stores local file to server.
- QUIT :** Gracefully closes control connection.